

# Vincent T. Cook

## MANUFACTURING & PRODUCTION

San Jose, CA | (805) 397-9594 | Vincent.Tait.Cook@gmail.com  
linkedin.com/in/vincent-cook-080790227

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Aerospace engineering senior with CAD assembly design and electric motor test-bench documentation experience. Apply structural and materials coursework, wind-tunnel testing, wing-model design, and project planning to manufacturing and production engineering.

## EDUCATION

San Jose State University | San Jose, CA  
B.S. Aerospace Engineering | Expected May 2027 | Enrolled Sep 2022

## TECHNICAL SKILLS

Design & documentation: SolidWorks, Autodesk Inventor, 3D CAD, assembly instructions, stress analysis  
Analysis & tools: simulation and modeling, CFD, thermal testing, MATLAB, Python, C++, Java

## ENGINEERING EXPERIENCE & PROJECTS

### SAE Spartan Racing

EV Powertrain Intern | Aug 2023–May 2025

- Developed 3D CAD models and assembly instructions for an electric motor performance test bench, defining how mechanical and electrical components fit together.
- Collaborated with teammates to align engineering and manufacturing practices and troubleshoot technical challenges.

### Hybrid Electric / Hydrogen Fuel-Cell Aircraft | SJSU Senior Project

Aerospace Engineering Team Member — In Progress | 2026–Present

- Contribute to wing-model design and project planning for a hybrid battery-electric and hydrogen fuel-cell model aircraft.
- Planned work includes CAD, CFD and thermodynamic analysis, aircraft construction/testing, and battery/fuel-system electronics integration.
- Planned scope also includes commercial-scale modeling; the team targets first flight in spring 2027.

### SJSU Robotics

Mechanical Designer | Aug 2024–Present

- Designed a transmitter mast integrating camera and sensor mounting features for a student robot.
- Used stress analysis to refine robot structure, reinforce high-stress regions, and select lighter materials for lower-load sections.

### BROADSWORD | Reusable Rocket Concept

CAD & Programming Subteam Lead | Sep–Dec 2023

- Created landing-leg CAD models for a reusable rocket concept and developed code producing trajectory-angle-versus-time plots.
- Delegated design and programming tasks, coordinated subteam work, and provided constructive technical feedback.

### NASA L'SPACE Mission Concept Academy

Mechanical Engineering Team Member — Educational Program | Fall 2026–Present

- Help plan scientific approaches and evaluate mechanical-component options for a robotic lunar surface payload mission concept.

## SELECTED COURSEWORK

- AE 20 | Motor CAD: Created high-level motor designs in introductory CAD coursework.
- AE 112 / AE 114 | Structures & materials: Studied structural dynamics and physical/material properties.
- AE 160 / AE 162 | Wind-tunnel labs: Used open wind-tunnel lab simulations to test wing airfoils and other objects.

## WORK EXPERIENCE

### The Tech Interactive | Educator I | 2024–2025

- Led hands-on STEM labs and design challenges, communicating procedures and technical concepts clearly.

### Self-Employed | Private Swim Instructor | 2023

- Coordinated multiple clients, scheduled lessons, and tailored instruction to individual needs.

### YMCA & Ventura Aquatics Center | Swim Instructor / Coach / Lifeguard | 2021–2024

- Balanced coaching and lifeguard responsibilities while adapting lessons for swimmers with varied abilities.